DATA DOCUMENTATION INITIATIVE

DDI 4 – The Potential as a Metadata Standard for Cross-Domain Research Data

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DDI 4 Initial Plan

- DDI 4 was started years ago with the ideas …
  - … to comprehend …
    - all features of all existing DDI specifications (DDI-C, DDI-L, Disco)
    - description of new data structures
    - description of new data sources beyond surveys
  - … to be driven by an UML model
  - … to have generated syntax representations
    - XML Schema for preservation and exchange
    - OWL /RDF for discovery and interoperability

Technical Core Features of DDI 4

- Model is a library of classes (templates)
  - Usage mainly in user-oriented subsets
    - Intended for specific applications
  - Architecture of DDI 4 can be understood as a kernel on which additional features and user-oriented views can be arranged on layers
- Specification as UML model
  - Advantages
    - Visual expression
    - Persistent model
    - Interoperable
      - With other models, other expressions can be derived from
- Derived syntax representations (XSD, OWL, others)

(Sub-)Sets of Metadata

- Specific concerns of common use cases define sets of metadata (subsets of the whole model)
- Examples:
  - Codebook
  - Instrument (survey and data collection types)
- Viewpoints of stakeholders (like archives) define these subsets
- Subsets are called here functional views

DDI 4 Status

- Still in development
- Public review of prototype (up to March 2019)
- Work on DDI 4 Core since beginning of 2019
  - DDI 4 Core is just a working name (proposals are DDI Cross-Domain, DDI Interoperable)

Background

- Recent developments in the research world are placing a greater emphasis on cross-domain integration of data, and data coming from new sources, some of it in unfamiliar forms (e.g., “big data,” social media, sensor data
- Social Science is no different from other domains – the DDI community is faced with a requirement for a more flexible ability to describe and manage, now available in a wider variety of forms

What is DDI 4 Core

- Large parts of DDI 4 are independent from the social science domain, especially in the areas of
  - Conceptual
    - the target entity or population, theoretical concepts
  - Data description
    - variable cascade which links conceptual definitions down to the instance variable
    - datum-based approach which enables the description of data in many forms (also data lakes and legacy forms)
  - Workflows (for process description)
    - Patterns of BPMN (Business Process Model and Notation)
- Use DDI to **D**ocument, **D**iscover and **I**nteroperate

Conceptual and Data Description

- Description of variables (variable cascade)
  - as they are used and re-used at different process stages
  - with a strong connection to a conceptual layer
- Traditional data structures
  - unit-record/rectangular data files
  - multi-dimensional/aggregate data
- Non-traditional data structures
  - event-history ("tall-skinny") data
  - other data formats (analysis of no-SQL is on-going)
- Description of classifications as used in description of data

Process description

- The ability to describe in detail the processes to which data have been subject
  - as an aspect of data provenance ("data lineage")
  - Data transformation and harmonization
  - (data capture and methodology are not described in detail, future task)

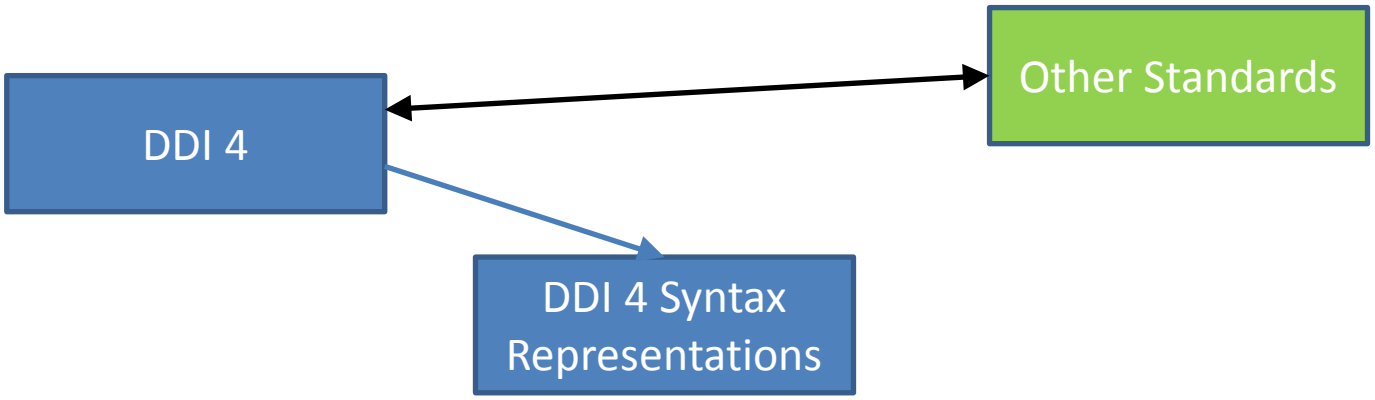
Alignment with other Metadata Standards

Because of the use of DDI 4 Core for cross-domain integration, and for other purposes, some key external standards have been selected as candidates for specific alignment

- GSIM (Generic Statistical Information Model, official statistics) as a higher-level model of core metadata
- GSBPM (Generic Statistical Business Process Model, official statistics) as a framework for process description
- DCAT (Data Catalog Vocabulary, W3C) for data discovery
- PROV Ontology (W3C) for provenance and data lineage
- Use of existing RDF vocabularies in the RDF syntax representation of the model is anticipated
  - Needed feature of alignment with standards/best practice in the Linked Data domain.

Interoperability

- UML model enables …
  - combination with other standards and models in other domains
  - generation of syntax representations
    - common in other domains
    - also future ones
  - further processing for other purposes



Technical Deliverables

- UML model for …
  - combination with other models and
  - generation of other program syntaxes
  - Expressed in Canonical XMI (portable flavor)
    - Supported by UML tools (further processing)
- XML Schema for preservation and exchange
  - for use in XML implementations
- OWL for discovery in the Web of Linked Data
  - for use in RDF implementations
  - with supporting validation resources (i.e., ShEx)

OBJECT MANAGEMENT GROUP®

Documentation Deliverables

- Complete documentation of the syntax representations for XML Schema and OWL
- Complete documentation of the classes and relationships in the UML model
- High-level documentation, including an explanation of design, features, and scope, and introductory material for the major features of the model (the variable cascade, the datum-based data description, and the process model for describing provenance/lineage)
- Documentation for implementation, alignment, and/or integration with external standards (PROV-O, DCAT, GSBPM, GSIM)

Variable Cascade

From concept to data value

- Concept
  - The concept layer provides an overarching concept from which a variable is defined. Examples include gender, income, occupation, and wages.
- Conceptual Variable
  - The conceptual variable includes the concept as the meaning of the variable, the associated unit type, and the categories the variable may assign. Note, these categories might be enumerated or described by a set of rules. The categories include processing or sentinel categories. These include reasons why data might be missing or incomplete.
- Represented Variable
  - The represented variable includes everything already specified for the conceptual variable plus the way the categories are represented, the intended datatype, a unit of measure if applicable, and any precision (e.g., number of decimal places) if applicable.
- Instance Variable
  - The instance variable includes everything already specified for the represented variable plus the representations for any sentinel (processing) categories.
- Value Mapping
  - The value mapping provides the means to find and retrieve data corresponding to the instance variable defined above out of a record in a file. It includes the specification of a physical datatype.

Each of the upper four layers may be reused to help specify layers underneath.

Three similar datasets - code representation

dataset1

name	height	birthdate	martialstatus
John	178	1998-09-02	S
Gill	200	1934-06-12	M
Alice	182	1922-12-24	M

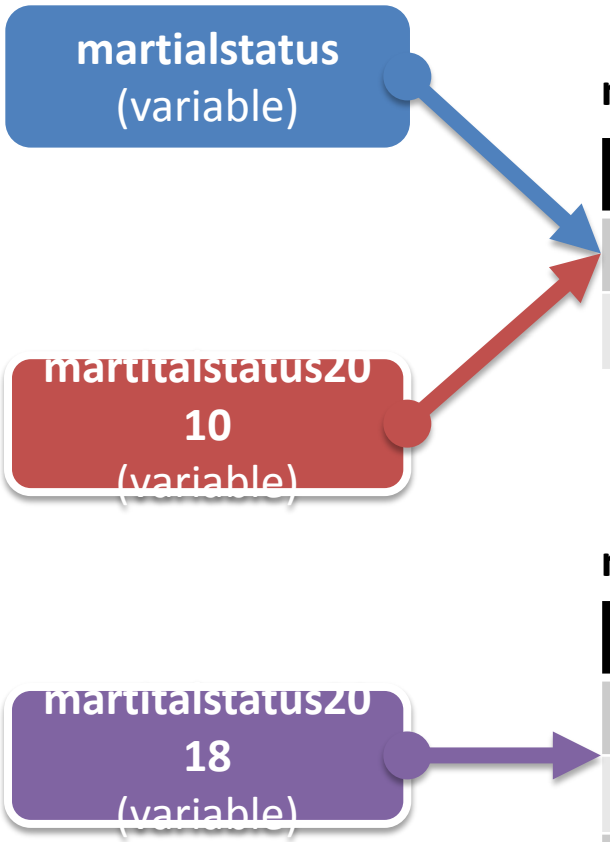
dataset2

firstname	personheight	dateofbirth	maritalstatus2010
Bob	70	1995-09-02	S
Lars	76	1954-06-21	M
Gerald	66	1972-11-23	S

dataset3

firstname	imperialheight	dateofbirth	maritalstatus2018
Lisa	69	1995-09-02	S
Bart	75	1954-06-21	M
Homer	68	1972-11-23	D

Variables re-using sets of codes



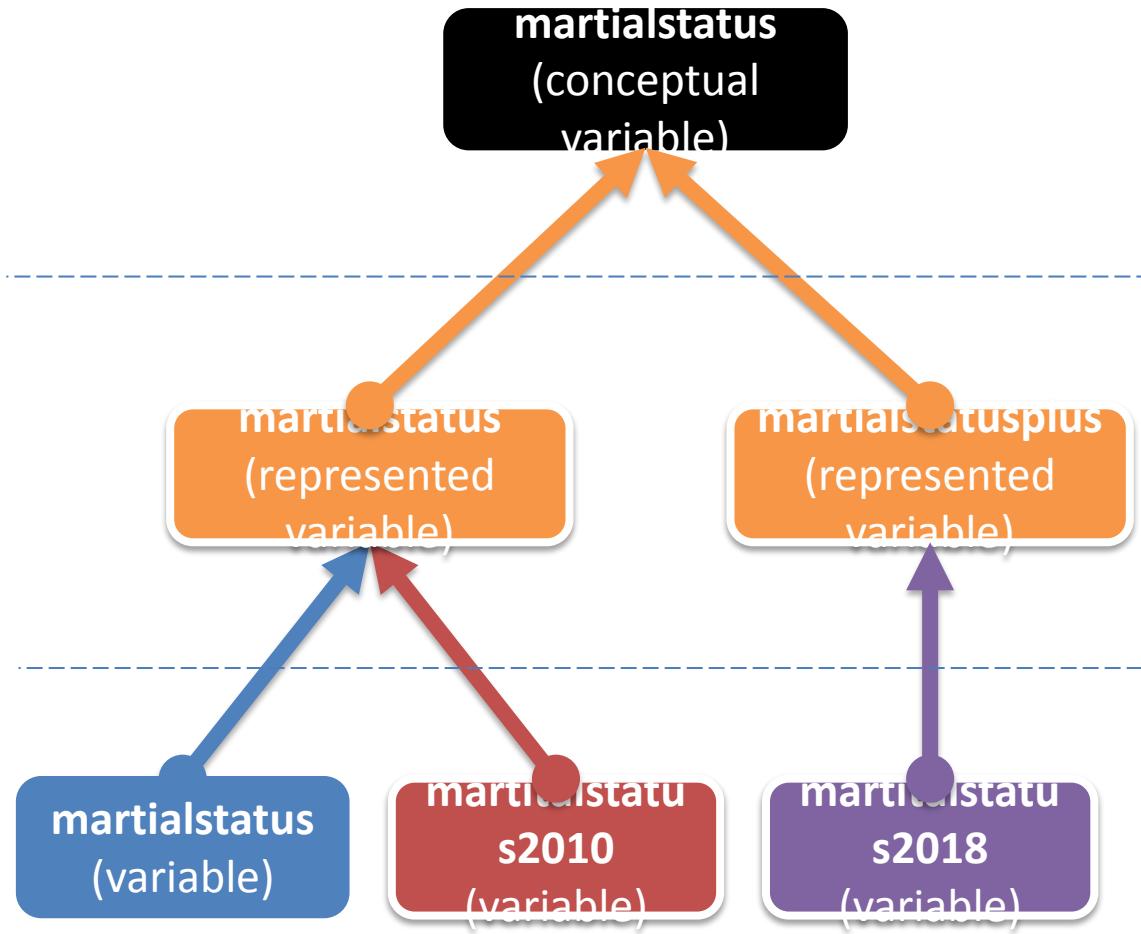
maritalstatus codes

code	category
S	Single
M	Married

maritalstatusplus

code	category
S	Single
M	Married
D	Divorced

Documenting comparabilities among variables



Conceptual variable
Common variable specification without a representation

Represented variable
Common variable specification with a *code representation*

Variable
Variable specification within a dataset context

Benefits of the variable cascade structure

- Specify comparability across variables allows comparison across datasets
- Structure for facilitating harmonization across datasets / studies
- Variable names and representations can change
- Documenting changes over time
- Planning for future data collection to ensure comparability

Interoperability of Metadata Standards in Cross-Domain Science, Health, and Social Science Applications

- Mutual influence between DDI and recent workshop in Dagstuhl (2nd workshop planned in October)
- Better understanding of how metadata specifications can be aligned to support cross-discipline (or cross domain) data integration and analysis
- Specific case studies in infectious disease, disaster risk and resilient cities

Dimensions in the Interdisciplinary Use of Data and Metadata

- From high-level to detail:
  - How to support discovery of data on the basis of a high-level description, and possibly also on more details like concepts related to observations and variables?
- From domain-specific to cross-domain:
  - How can domain-specific data/metadata be transformed in a way that cross-domain approaches regarding discovery and analysis can be supported?

Issues in Discovery and Assessment of Datasets

- Not precise enough:
  - General keywords, subject headings – not tied to a well-governed terminology
  - Abstract
- Not processable by program:
  - Link from metadata to data often obscured by an intermediate „landing page“

Possible Solutions

- More precise semantics with tags on
  - variables, properties or dimensions reported in the dataset
  - the target entity or population that a dataset describes
  - the sample(s) of that entity used in the specific investigation
  - provide summary statistics to indicate thematic, geographic and temporal scope and resolution of a dataset
- Question: How can i.e. DCAT support these in combination with domain-specific standards?

DCAT Profile for DDI

- Example for handover from generic level to specific level
  - See presentation „Harmonising data description across the social and natural sciences", Simon Cox et al.
- Profiles of DCAT are required for specific applications and disciplines (from DCAT revised edition 2018)
- Machine-readable specifications of application profiles need to be easily publishable, and optimize re-use of existing specifications (from ongoing DXWG work, issue 75)

Results of Dagstuhl 2018

All still in the works

- Open Standards and the FAIR Principles: Three Use Cases
- The Value of Data Science for the Implementation of the Sendai Framework for Disaster Risk Reduction 2015–2030
- Challenges and Opportunities in the Use of Data Integration to Fight Infectious Diseases
- Urban Data Integration for Resilient Planning: Methodological Considerations
- Web Research Metadata Profile for Data Discovery and Re-use
- Data Science for Data Discovery and Integration: Possible Approaches

DDI 4 Outlook

- Work on DDI 4 Core is going on
  - Comprehensive and iterative approach on modeling, syntax representation, and testing
- Plan of final deliverable ready for review at the end of December 2019
- Additional extensions and views planned for 2020+ like …
  - data capture, codebook, and qualitative

Credits

- DDI Alliance Working Groups
  - DDI Moving Forward Project, especially …
    - Data Description
    - Modeling / MRT (Modeling, Representation, Testing)
  - Technical Committee
- Dagstuhl Workshop participants of
    - Interoperability of Metadata Standards in Cross-Domain Science, Health, and Social Science Applications, Oct 2018
    - Train-the-trainers, Sep 2018

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